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Split-step balanced θ-method for stochastic differential equations under non-global Lipschitz conditions☆

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Abstract

In this paper, the split-step balanced θ-method (SSBT) has been presented for solving stochastic differential equations (SDEs) under non-global Lipschitz conditions. The boundedness and convergence of the numerical solution have been studied in mean-square sense. Furthermore, under some condition it is proved that the SSBT scheme can preserve the exponential mean-square stability of the exact solution when $\theta \in [1/2, 1]$ for every stepsize $h > 0$. Numerical examples verify the theoretical findings.

Key words: nonlinear problems, the balanced method, strong convergence, exponential stability, mean-square contraction
MSC subject classifications: 65C30, 65L20, 65L03, 60H35

1. Introduction

Stochastic differential equations (SDEs) have been widely used in many branches of science and industry. The convergence and stability of numerical methods are well studied for SDEs with globally Lipschitz continuous coefficients [1, 15, 18, 19, 23, 28, 30].

However, the coefficients of some reliable complex systems in many application fields, such as economics, physics, engineering, etc., are nonlinear and violate the global Lipschitz conditions. When solving such nonlinear systems, explicit schemes generally are neither stable nor convergent [7, 8, 12, 15]. To deal with the problems, some kinds of numerical schemes have been constructed, e.g., the tamed/balanced schemes [2, 11, 26, 31, 32], the truncated schemes [16, 21], the projected explicit Itô-Taylor method [9], etc.

Explicit schemes usually have a simple structure and cheap computational cost, but when solving strong nonlinear/stiff problems, implicit schemes are more efficient than explicit schemes due to less restriction on step-size. The semi-implicit Euler-Maruyama scheme (the backward Euler scheme when $\theta = 1$) is very popular, and its convergence and stability have been well studied for SDEs under non-global Lipschitz conditions. For example, considering the SDE with one-sided Lipschitz continuous drift coefficient and globally Lipschitz continuous diffusion coefficient [10] and the SDE under the local Lipschitz conditions and monotone condition [20]. In [17] and [33], it was proved that the backward Euler method is almost surely exponential stable for solving some highly nonlinear

☆This work was partially supported by the NSF of China (No. 11671083 and No. 12071073) and the Fundamental Research Funds for the Central Universities (No. 2242019S20038).
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SDEs. Moreover, the convergence and stability of θ-Milstein scheme and double-implicit Milstein schemes are studied in [29, 34]. A fully implicit scheme was also proposed in [27].

Both the balanced scheme and the splitting technique are powerful for solving stiff problems. For SDEs with globally Lipschitz continuous coefficients, some split-step schemes and split-step balanced schemes have been presented, which have shown good stability behavior, see [1, 5, 24, 25, 28]. There are only a few works on split-step schemes for SDEs under non-globally Lipschitz conditions. In [4, 13], convergence of the split-step Euler-type schemes were studied for SDEs under non-global Lipschitz conditions. Higham et al. proved that the backward Euler-Maruyama and the split-step backward Euler-Maruyama can preserve the exponential mean-square stability of nonlinear SDEs [6], and Huang showed that when $\theta \in (\frac{1}{2}, 1]$, the split-step θ scheme could preserve the exponential mean-square stability of exact solution for SDEs with a coupled non-global Lipschitz conditions [3].

In this work, we consider the following autonomous SDE in Itô sense:

$$\begin{cases} dX(t) = f(X(t))dt + g(X(t))dW(t), \quad 0 < t \leq T, \\ X(0) = X_0, \end{cases} \tag{1.1}$$

where $f : \mathbb{R}^n \mapsto \mathbb{R}^n$ and $g : \mathbb{R}^n \mapsto \mathbb{R}^{n\times d}$, $W(t)$ is a d-dimensional standard Brownian motion, $X(0)$ is independent of $W(t)$.

Our work aims to construct a split-step balanced θ-scheme (SSBT) for the SDE (1.1) with a series of non-global Lipschitz conditions (see Assumption 2.1), and to study its convergence and exponential stability in strong sense. The main contribution is to prove that under some non-global Lipschitz conditions the presented scheme is of a half order convergence in mean-square sense and can preserve the exponential stability of the exact solution. To the best of the authors' knowledge, this variant of balanced schemes has not been considered before under the non-global Lipschitz conditions. Numerical examples verify the theoretical predictions and show that the proposed scheme is efficient.

The remainder of the paper is organized as follows. In Section 2 we give necessary notations and some assumptions on the initial value and coefficients of the SDE (1.1). In Section 3, we present the SSBT scheme and prove the boundedness and strong convergence of the numerical solution. The contraction and exponential stability of the SSBT scheme are considered in Section 4. Some numerical examples are given in Section 5 to verify the theoretical findings on both convergence rate and exponential stability of the SSBT scheme in mean-square sense. Finally, we give some concluding remarks in Section 6.

2. Notations and assumptions

Let $(\Omega, \mathcal{F}, P)$ be a complete probability space, where $\mathcal{F}_t,\ t_0 \leq t \leq T$, is a nondecreasing family of σ-subalgebras of $\mathcal{F}$. The coefficients in Eq. (1.1) are Borel measurable. We will use $\langle x, y\rangle$ and $|\cdot|$ to denote the inner product and the Euclidean norm of vectors x, y in n-dimensional Euclidean space $\mathbb{R}^n$, respectively. Also, $|\cdot|$ denotes the norm of matrix in $\mathbb{R}^{n\times d}$. Throughout the paper, the capital letter K denotes a dynamic constant.

In this work, we make the following assumption.

Assumption 2.1.

(i) For all $p \geq 1$, the initial condition satisfies

$$\mathbb{E}[|X(0)|^{2p}] \leq K < \infty. \tag{2.1}$$

(ii) For a sufficiently large $p_0 \geq 1$, there is a constant $c_1 \geq 0$ such that $\forall\ x, y \in \mathbb{R}^n$

$$\langle x-y, f(x)-f(y)\rangle + \frac{2p_0-1}{2}|g(x)-g(y)|^2 \leq c_1|x-y|^2. \tag{2.2}$$

(iii) There exist $c_2 \geq 0$ and $\kappa \geq 1$ such that $\forall\ x, y \in \mathbb{R}^n$

$$|f(x)-f(y)|^2 \leq c_2(1+|x|^{2\kappa-2}+|y|^{2\kappa-2})|x-y|^2. \tag{2.3}$$

We can derive the following corollary from Assumption 2.1 easily.

Corollary 2.2. *Suppose that Assumption 2.1 holds, then there exist constants c_3, c_4, c_5 such that $\forall\ x, y \in \mathbb{R}^n$*

$$\langle x, f(x)\rangle + \frac{2p_0-3}{2}|g(x)|^2 \leq c_3(1+|x|^2), \tag{2.4}$$

$$|f(x)|^2 \leq c_4(1+|x|^{2\kappa}), \tag{2.5}$$

$$|g(x)|^2 \leq c_5(1+|x|^{2\kappa}). \tag{2.6}$$

Remark 2.3. *The inequalities (2.1) and (2.4) can guarantee the boundedness of the analytical solution of the SDE (1.1) [14]*

$$\mathbb{E}[|X(t)|^{2p}] \leq K\left(1+\mathbb{E}[|X_0|^{2p}]\right), \qquad 1 \leq p \leq p_0-1. \tag{2.7}$$

3. Boundeness and convergence of the SSBT scheme

This section will introduce the split-step balanced θ-scheme (SSBT) for the SDE (1.1) and prove the boundedness and convergence of the numerical solution. Take some integer N and let $h = \frac{T}{N}$. Denote $t_k = kh,\ 0 \leq k \leq N$, X_k to be approximation of $X(t_k)$ for $1 \leq k \leq N$, and $X_0 = X(0)$.

The SSBT scheme of the SDE (1.1) has the form

$$\begin{cases} X_k^* = X_k + \theta f(X_k^*)h, \\ X_{k+1} = X_k + \dfrac{f(X_k^*)h + g(X_k^*)\xi_k\sqrt{h}}{1+|f(X_k^*)|h+|g(X_k^*)\xi_k|\sqrt{h}}, \quad 0 \leq k \leq N-1, \end{cases} \tag{3.1}$$

where ξ_k are d-dimensional vectors and every component of ξ_k is Gaussian $\mathcal{N}(0,1)$ i.i.d random variable.

For convenience, we denote $B(X_k^*, \xi_k) := (1+|f(X_k^*)|h+|g(X_k^*)\xi_k|\sqrt{h})^{-1}$.

Lemma 3.1. *Suppose that Assumption 2.1 holds. For the conditional expectation*

$$A(X_k, X_k^*, \xi_k) := \mathbb{E}[\langle X_k, f(X_k^*)h\rangle B(X_k^*, \xi_k) + \frac{2p-1}{2}\left(|f(X_k^*)|^2 h^2 + |g(X_k^*)\xi_k|^2 h\right) B(X_k^*, \xi_k)|\mathcal{F}_{t_k}],$$

where $1 \leq p \leq p_0-1$, there exist a constant $K > 0$ independent of h and k, such that

$$A(X_k, X_k^*, \xi_k) \leq Kh\left(1+|X_k^*|^2+|X_k^*|^{3\kappa}h^{\frac{1}{2}}\right).$$

Proof Noting that X_k, X_k^* are $\mathcal{F}_{t_k}$-measurable and $B(X_k^*, \xi_k) \leq 1$, we have

$$\begin{aligned}
&A(X_k, X_k^*, \xi_k) \\
\leq&\mathbb{E}[\langle X_k, f(X_k^*)h\rangle B(X_k^*, \xi_k) + \frac{2p-1}{2}|g(X_k^*)|^2 \xi_k^2 h B(X_k^*, \xi_k)|\mathcal{F}_{t_k}] + \frac{2p-1}{2}|f(X_k^*)|^2 h^2 \\
=&\mathbb{E}[(\langle X_k - X_k^*, f(X_k^*)h\rangle + \langle X_k^*, f(X_k^*)h\rangle) B(X_k^*, \xi_k) + \frac{2p-1}{2}|g(X_k^*)|^2 \xi_k^2 h B(X_k^*, \xi_k)|\mathcal{F}_{t_k}] \\
&+ \frac{2p-1}{2}|f(X_k^*)|^2 h^2 \\
\leq&\mathbb{E}[\theta|f(X_k^*)|^2 h^2 B(X_k^*, \xi_k) + \left(\langle X_k^*, f(X_k^*)h\rangle + \frac{2p-1}{2}h|g(X_k^*)|^2\right) B(X_k^*, \xi_k) \\
&+ \frac{2p-1}{2}h|g(X_k^*)|^2(\xi_k^2-1)B(X_k^*, \xi_k)|\mathcal{F}_{t_k}] + \frac{2p-1}{2}|f(X_k^*)|^2 h^2.
\end{aligned}$$

By using the estimation (2.4), we obtain

$$\begin{aligned}A(X_k, X_k^*, \xi_k) \leq & \frac{2p-1}{2} h \left|g(X_k^*)\right|^2 \mathbb{E}[(\xi_k^2-1)B(X_k^*,\xi_k)\big|\mathcal{F}_{t_k}] + \theta \left|f(X_k^*)\right|^2 h^2 \\ & + c_3(1+|X_k^*|^2)h + \frac{2p-1}{2}\left|f(X_k^*)\right|^2 h^2.\end{aligned}$$

Since $\mathbb{E}[\xi_k^2-1]=0$ and ξ_k is independent of $\mathcal{F}_{t_k}$, it follows that

$$\begin{aligned}\mathbb{E}[(\xi_k^2-1)B(X_k^*,\xi_k)\big|\mathcal{F}_{t_k}] &= \mathbb{E}[(\xi_k^2-1)B(X_k^*,\xi_k) - (\xi_k^2-1)\big|\mathcal{F}_{t_k}] \\ &= -\mathbb{E}[(\xi_k^2-1)(|f(X_k^*)|h + |g(X_k^*)\xi_k|\sqrt{h})B(X_k^*,\xi_k)\big|\mathcal{F}_{t_k}] \\ &\leq \mathbb{E}[|\xi_k^2-1|(|f(X_k^*)|h + |g(X_k^*)\xi_k|\sqrt{h})\big|\mathcal{F}_{t_k}] \\ &\leq K(|f(X_k^*)|h + |g(X_k^*)|\sqrt{h}).\end{aligned}$$

According to (2.5) and (2.6), we have

$$\begin{aligned}A(X_k, X_k^*, \xi_k) &\leq Kh + K|X_k^*|^2 h + Kh|g(X_k^*)|^2(|f(X_k^*)|h + \sqrt{h}|g(X_k^*)|) + Kh^2 + K|X_k^*|^{2\kappa}h^2 \\ &\leq Kh(1+|X_k^*|^2 + |X_k^*|^{2\kappa}h + |X_k^*|^{3\kappa}h^{\frac{1}{2}}) \\ &\leq Kh(1+|X_k^*|^2 + |X_k^*|^{3\kappa}h^{\frac{1}{2}}).\end{aligned}$$

□

Theorem 3.2. *Suppose that Assumption 2.1 holds. If $\theta h < \frac{1}{2c_3}$, then for the SSBT scheme (3.1) we have*

$$\mathbb{E}[|X_k^*|^{2p}] \leq K(1+\mathbb{E}[|X_0|^{2p\beta}]), \quad \mathbb{E}[|X_k|^{2p}] \leq K(1+\mathbb{E}[|X_0|^{2p\beta}]), \quad 0 \leq k \leq N, \tag{3.2}$$

where $K>0$, $\beta \geq 1$ are constants independent of h, k, and $1 \leq p \leq \frac{p_0-1}{4(3\kappa-2)} - \frac{1}{2}$.

Proof We observe from the SSBT scheme (3.1) that

$$|X_{k+1}| \leq |X_k| + 1 \leq |X_0| + k + 1. \tag{3.3}$$

Let $R>0$ be a sufficiently large number. Introduce the sets

$$\tilde{\Omega}_{R,k} := \{\omega : |X_l(\omega)| \leq R,\ l = 0,1,\ldots,k\}$$

and their complements $\tilde{\Lambda}_{R,k}$. Denote E is an event set and $\chi_E(\omega)=1$, when $\omega \in E$; $\chi_E(\omega)=0$, otherwise.

We first prove the theorem for integer $p \geq 1$. We have

$$\begin{aligned}\mathbb{E}[\chi_{\tilde{\Omega}_{R,k+1}}(\omega)|X_{k+1}|^{2p}] \leq & \mathbb{E}[\chi_{\tilde{\Omega}_{R,k}}(\omega)|X_{k+1}|^{2p}] \\ = & \mathbb{E}[\chi_{\tilde{\Omega}_{R,k}}(\omega)|X_{k+1}-X_k+X_k|^{2p}] \\ \leq & \mathbb{E}[\chi_{\tilde{\Omega}_{R,k}}(\omega)|X_k|^{2p-2}\left(2p\langle X_k, X_{k+1}-X_k\rangle + p(2p-1)|X_{k+1}-X_k|^2\right)] \\ & + K\sum_{l=3}^{2p}\mathbb{E}[\chi_{\tilde{\Omega}_{R,k}}(\omega)|X_k|^{2p-l}|X_{k+1}-X_k|^l] + \mathbb{E}[\chi_{\tilde{\Omega}_{R,k}}(\omega)|X_k|^{2p}].\end{aligned} \tag{3.4}$$

Note that $\chi_{\tilde{\Omega}_{R,k}}(\omega)$ and X_k are $\mathcal{F}_{t_k}$-measurable. According to

$$\mathbb{E}[B(X_k^*,\xi_k)g(X_k^*)\xi_k\sqrt{h}\big|\mathcal{F}_{t_k}] = 0, \tag{3.5}$$

and Lemma 3.1, the first term in the right-hand side of (3.4) yields

$$
\begin{aligned}
&\mathbb{E}[\chi_{\tilde{\Omega}_{R,k}}(\omega)|X_k|^{2p-2}\big(2p\langle X_k, X_{k+1}-X_k\rangle + p(2p-1)|X_{k+1}-X_k|^2\big)] \\
&= 2p\mathbb{E}[\chi_{\tilde{\Omega}_{R,k}}(\omega)|X_k|^{2p-2}\mathbb{E}[\langle X_k, (f(X_k^*)h + g(X_k^*)\xi_k\sqrt{h})B(X_k^*,\xi_k)\rangle \\
&\quad + \frac{2p-1}{2}\big|(f(X_k^*)h + g(X_k^*)\xi_k\sqrt{h})\big|^2 B^2(X_k^*,\xi_k)\big|\mathcal{F}_{t_k}]] \\
&\le K\mathbb{E}[\chi_{\tilde{\Omega}_{R,k}}(\omega)|X_k|^{2p-2}A(X_k, X_k^*, \xi_k)] \\
&\le Kh\mathbb{E}[\chi_{\tilde{\Omega}_{R,k}}(\omega)|X_k|^{2p-2}(1+|X_k^*|^2+|X_k^*|^{3\kappa}h^{\frac{1}{2}})].
\end{aligned} \tag{3.6}
$$

Now consider the second term in the right-hand side of (3.4). Using (2.5), (2.6) and $B(X_k^*,\xi_k)\le 1$, it follows that

$$
\begin{aligned}
\mathbb{E}[\chi_{\tilde{\Omega}_{R,k}}(\omega)|X_k|^{2p-l}|X_{k+1}-X_k|^l] &\le K\mathbb{E}[\chi_{\tilde{\Omega}_{R,k}}(\omega)|X_k|^{2p-l}\big(h^l|f(X_k^*)|^l + h^{\frac{l}{2}}|g(X_k^*)|^l|\xi_k|^l\big)] \\
&\le K\mathbb{E}[\chi_{\tilde{\Omega}_{R,k}}(\omega)|X_k|^{2p-l}h^{\frac{l}{2}}\left(1+|X_k^*|^{l\kappa}\right)].
\end{aligned} \tag{3.7}
$$

Substituting (3.6) and (3.7) into (3.4), we obtain

$$
\begin{aligned}
&\mathbb{E}[\chi_{\tilde{\Omega}_{R,k+1}}(\omega)|X_{k+1}|^{2p}] \\
\le&\mathbb{E}[\chi_{\tilde{\Omega}_{R,k}}(\omega)|X_k|^{2p}] + Kh\mathbb{E}[\chi_{\tilde{\Omega}_{R,k}}(\omega)|X_k|^{2p-2}\big(1+|X_k^*|^2+|X_k^*|^{3\kappa}h^{\frac{1}{2}}\big)] \\
&+ K\sum_{l=3}^{2p}\mathbb{E}[\chi_{\tilde{\Omega}_{R,k}}(\omega)|X_k|^{2p-l}h^{\frac{l}{2}}\left(1+|X_k^*|^{l\kappa}\right)] \\
=&\mathbb{E}[\chi_{\tilde{\Omega}_{R,k}}(\omega)|X_k|^{2p}] + Kh\mathbb{E}[\chi_{\tilde{\Omega}_{R,k}}(\omega)|X_k|^{2p-2}] + Kh\mathbb{E}[\chi_{\tilde{\Omega}_{R,k}}(\omega)|X_k|^{2p-2}|X_k^*|^2] \\
&+ Kh\mathbb{E}[\chi_{\tilde{\Omega}_{R,k}}(\omega)|X_k|^{2p-2}|X_k^*|^{3\kappa}h^{\frac{1}{2}}] + K\sum_{l=3}^{2p}\mathbb{E}[\chi_{\tilde{\Omega}_{R,k}}(\omega)|X_k|^{2p-l}h^{\frac{l}{2}}] \\
&+ Kh\sum_{l=3}^{2p}\mathbb{E}[\chi_{\tilde{\Omega}_{R,k}}(\omega)|X_k|^{2p-l}|X_k^*|^{l\kappa}h^{\frac{l}{2}-1}].
\end{aligned} \tag{3.8}
$$

According to Young's inequality

$$
\mathbb{E}[\chi_{\tilde{\Omega}_{R,k}}(\omega)|X_k|^{2p-2}|X_k^*|^2] \le \mathbb{E}[\chi_{\tilde{\Omega}_{R,k}}(\omega)\left(\frac{p-1}{p}|X_k|^{2p} + \frac{1}{p}|X_k^*|^{2p}\right)].
$$

Then

$$
\mathbb{E}[\chi_{\tilde{\Omega}_{R,k}}(\omega)|X_k|^{2p-2}|X_k^*|^2] \le K\left(\mathbb{E}[\chi_{\tilde{\Omega}_{R,k}}(\omega)|X_k|^{2p}] + \mathbb{E}[\chi_{\tilde{\Omega}_{R,k}}(\omega)|X_k^*|^{2p}]\right).
$$

Similarly, we can obtain

$$
\begin{aligned}
\mathbb{E}[\chi_{\tilde{\Omega}_{R,k}}(\omega)|X_k|^{2p-2}|X_k^*|^{3\kappa}h^{\frac{1}{2}}] &\le K\left(\mathbb{E}[\chi_{\tilde{\Omega}_{R,k}}(\omega)|X_k|^{2p-2+3\kappa}h^{\frac{1}{2}}] + \mathbb{E}[\chi_{\tilde{\Omega}_{R,k}}(\omega)|X_k^*|^{2p-2+3\kappa}h^{\frac{1}{2}}]\right), \\
\mathbb{E}[\chi_{\tilde{\Omega}_{R,k}}(\omega)|X_k|^{2p-l}|X_k^*|^{l\kappa}h^{\frac{l}{2}-1}] &\le K\left(\mathbb{E}[\chi_{\tilde{\Omega}_{R,k}}(\omega)|X_k|^{2p+l(\kappa-1)}h^{\frac{l}{2}-1}] + \mathbb{E}[\chi_{\tilde{\Omega}_{R,k}}(\omega)|X_k^*|^{2p+l(\kappa-1)}h^{\frac{l}{2}-1}]\right).
\end{aligned}
$$

Therefore, the inequality (3.8) becomes

$$\begin{aligned}\mathbb{E}[\chi_{\tilde{\Omega}_{R,k+1}}(\omega)|X_{k+1}|^{2p}] \leq & \mathbb{E}[\chi_{\tilde{\Omega}_{R,k}}(\omega)|X_k|^{2p}] + K\sum_{l=2}^{2p}\mathbb{E}[\chi_{\tilde{\Omega}_{R,k}}(\omega)|X_k|^{2p-l}h^{\frac{l}{2}}] + Kh\mathbb{E}[\chi_{\tilde{\Omega}_{R,k}}(\omega)|X_k|^{2p}] \\ & + Kh^{\frac{3}{2}}\mathbb{E}[\chi_{\tilde{\Omega}_{R,k}}(\omega)|X_k|^{2p-2+3\kappa}] + Kh\sum_{l=3}^{2p}\mathbb{E}[\chi_{\tilde{\Omega}_{R,k}}(\omega)|X_k|^{2p+l(\kappa-1)}h^{\frac{l}{2}-1}] \\ & + Kh\mathbb{E}[\chi_{\tilde{\Omega}_{R,k}}(\omega)|X_k^*|^{2p}] + Kh^{\frac{3}{2}}\mathbb{E}[\chi_{\tilde{\Omega}_{R,k}}(\omega)|X_k^*|^{2p-2+3\kappa}] \\ & + Kh\sum_{l=3}^{2p}\mathbb{E}[\chi_{\tilde{\Omega}_{R,k}}(\omega)|X_k^*|^{2p+l(\kappa-1)}h^{\frac{l}{2}-1}].\end{aligned}$$

By Young's inequality, from the second term of the right hand side of the above inequality we get

$$\sum_{l=2}^{2p}\mathbb{E}[\chi_{\tilde{\Omega}_{R,k}}(\omega)|X_k|^{2p-l}h^{\frac{l}{2}}] \leq Kh(1+\mathbb{E}[\chi_{\tilde{\Omega}_{R,k}}(\omega)|X_k|^{2p}]). \tag{3.9}$$

Choosing $R = R(h) = h^{-\frac{1}{6\kappa-4}}$, we can obtain that for $l = 3, \ldots, 2p$

$$\chi_{\tilde{\Omega}_{R(h),k}}(\omega)|X_k|^{2p-2+3\kappa}h^{\frac{l}{2}-1} \leq \chi_{\tilde{\Omega}_{R(h),k}}(\omega)|X_k|^{2p}, \tag{3.10}$$

$$\chi_{\tilde{\Omega}_{R(h),k}}(\omega)|X_k|^{2p+l(\kappa-1)}h^{\frac{l}{2}-1} \leq \chi_{\tilde{\Omega}_{R(h),k}}(\omega)|X_k|^{2p}. \tag{3.11}$$

Thus, it follows from (3.9)-(3.11) that

$$\begin{aligned}\mathbb{E}[\chi_{\tilde{\Omega}_{R(h),k+1}}(\omega)|X_{k+1}|^{2p}] \leq & \mathbb{E}[\chi_{\tilde{\Omega}_{R(h),k}}(\omega)|X_k|^{2p}] + Kh\mathbb{E}[\chi_{\tilde{\Omega}_{R(h),k}}(\omega)|X_k|^{2p}] + Kh \\ & + Kh\mathbb{E}[\chi_{\tilde{\Omega}_{R(h),k}}(\omega)|X_k^*|^{2p}] + Kh^{\frac{3}{2}}\mathbb{E}[\chi_{\tilde{\Omega}_{R(h),k}}(\omega)|X_k^*|^{2p-2+3\kappa}] \\ & + Kh\sum_{l=3}^{2p}\mathbb{E}[\chi_{\tilde{\Omega}_{R(h),k}}(\omega)|X_k^*|^{2p+l(\kappa-1)}h^{\frac{l}{2}-1}].\end{aligned} \tag{3.12}$$

By the SSBT scheme (3.1), we can have the following estimation.

$$\mathbb{E}[\chi_{\tilde{\Omega}_{R(h),k}}(\omega)|X_k^*|^{2p}] \leq K\mathbb{E}[\chi_{\tilde{\Omega}_{R(h),k}}(\omega)|X_k|^{2p}] + Kh^p. \tag{3.13}$$

In fact, by $X_k = X_k^* - \theta f(X_k^*)h$ and $\theta h < \frac{1}{2c_3}$, we have

$$\begin{aligned}|X_k|^2 &= |X_k^*|^2 - 2\theta h\langle X_k^*, f(X_k^*)\rangle + \theta^2h^2|f(X_k^*)|^2 \\ &\geq (1-2c_3\theta h)|X_k^*|^2 - 2\theta c_3 h.\end{aligned}$$

It follows that

$$(1-2c_3\theta h)^p|X_k^*|^{2p} \leq 2^{p-1}\left(|X_k|^{2p} + (2\theta c_3 h)^p\right),$$

i.e.

$$|X_k^*|^{2p} \leq K\left(|X_k|^{2p} + h^p\right), \tag{3.14}$$

and (3.13) holds.

Substituting inequality (3.13) into (3.12) we can get

$$\begin{aligned}\mathbb{E}[\chi_{\tilde{\Omega}_{R(h),k+1}}(\omega)|X_{k+1}|^{2p}] \leq & \mathbb{E}[\chi_{\tilde{\Omega}_{R(h),k}}(\omega)|X_k|^{2p}] + Kh\mathbb{E}[\chi_{\tilde{\Omega}_{R(h),k}}(\omega)|X_k|^{2p}] + Kh \\ & + Kh\mathbb{E}[\chi_{\tilde{\Omega}_{R(h),k}}(\omega)|X_k|^{2p}] + Kh^{\frac{3}{2}}\mathbb{E}[\chi_{\tilde{\Omega}_{R(h),k}}(\omega)|X_k|^{2p-2+3\kappa}] \\ & + Kh\sum_{l=3}^{2p}\mathbb{E}[\chi_{\tilde{\Omega}_{R(h),k}}(\omega)|X_k|^{2p+l(\kappa-1)}h^{\frac{l}{2}-1}].\end{aligned}$$

Utilizing (3.10) and (3.11) again, it is obtained that

$$\mathbb{E}[\chi_{\tilde{\Omega}_{R(h),k+1}}(\omega)|X_{k+1}|^{2p}] \leq \mathbb{E}[\chi_{\tilde{\Omega}_{R(h),k}}(\omega)|X_k|^{2p}] + Kh\mathbb{E}[\chi_{\tilde{\Omega}_{R(h),k}}(\omega)|X_k|^{2p}] + Kh. \tag{3.15}$$

According to Gronwall's inequality, we can get

$$\mathbb{E}[\chi_{\tilde{\Omega}_{R(h),k}}(\omega)|X_k|^{2p}] \leq K(1 + \mathbb{E}[|X_0|^{2p}]). \tag{3.16}$$

Next we focus on $\mathbb{E}[\chi_{\tilde{\Lambda}_{R,k}}(\omega)|X_k|^{2p}]$. The estimation is almost similar as that in [27]. For completeness, we still show it as follows.

$$\chi_{\tilde{\Lambda}_{R,k}} = 1 - \chi_{\tilde{\Omega}_{R,k}} = 1 - \chi_{\tilde{\Omega}_{R,k-1}}\chi_{|X_k|\leq R} = \chi_{\tilde{\Lambda}_{R,k-1}} + \chi_{\tilde{\Omega}_{R,k-1}}\chi_{|X_k|>R} = \cdots = \sum_{l=0}^{k}\chi_{\tilde{\Omega}_{R,l-1}}\chi_{|X_l|>R},$$

where $\chi_{\tilde{\Omega}_{R,-1}} = 1$.

According to (3.3), (3.16), Markov's inequality and Cauchy-Schwarz's inequality, we have

$$\begin{aligned}
\mathbb{E}[\chi_{\tilde{\Lambda}_{R(h),k}}(\omega)|X_k|^{2p}] &= \sum_{l=0}^{k}\mathbb{E}[\chi_{\tilde{\Omega}_{R(h),l-1}}\chi_{|X_l|>R}|X_k|^{2p}] \\
&\leq \left(\mathbb{E}\left[|X_0| + k\right]^{4p}\right)^{\frac{1}{2}}\left(\sum_{l=0}^{k}\mathbb{E}[\chi_{\tilde{\Omega}_{R(h),l-1},|X_l|>R(h)}]\right)^{\frac{1}{2}} \\
&= \left(\mathbb{E}\left[|X_0| + k\right]^{4p}\right)^{\frac{1}{2}}\left(\sum_{l=0}^{k}P(\chi_{\tilde{\Omega}_{R(h),l-1}}|X_l| > R(h))\right)^{\frac{1}{2}} \\
&\leq \left(\mathbb{E}\left[|X_0| + k\right]^{4p}\right)^{\frac{1}{2}}\sum_{l=0}^{k}\frac{\left(\mathbb{E}[\chi_{\tilde{\Omega}_{R(h),l-1}}|X_l|^{2(2p+1)(6\kappa-4)}]\right)^{\frac{1}{2}}}{R(h)^{(2p+1)(6\kappa-4)}} \\
&\leq K\left(\mathbb{E}\left[|X_0| + k\right]^{4p}\right)^{\frac{1}{2}}\left(\mathbb{E}[1 + |X_0|^{2(2p+1)(6\kappa-4)}]\right)^{\frac{1}{2}}kh^{2p+1} \\
&\leq K\left(1 + \mathbb{E}[|X_0|^{4p+2(2p+1)(6\kappa-4)}]\right)^{\frac{1}{2}},
\end{aligned}$$

which together with (3.16), the inequality (3.2) is proved for integer $p \geq 1$. Then, by Jensen's inequality, (3.2) holds for non-integer p as well. By using (3.14), we can obtain

$$\mathbb{E}[|X_k^*|^{2p}] \leq K\left(1 + \mathbb{E}[|X_0|^{2p\beta}]\right).$$

□

The following theorem states the strong convergence of the SSBT scheme.

Theorem 3.3. *Let Assumption 2.1 hold, and $X(t)$ be the exact solution of Eq. (1.1), and X_k, $0 \leq k \leq N$ be the numerical solution produced by the SSBT scheme (3.1) with parameter $\theta \in [0,1]$ and step size $h = T/N$. If $\theta h < \frac{1}{2c_3}$, then there exist constants $K > 0$, $\gamma \geq 1$ independent of h, k, such that*

$$\left(\mathbb{E}[|X(t_k) - X_k|^{2p}]\right)^{\frac{1}{2p}} \leq K\left(1 + \mathbb{E}[|X_0|^{2\gamma p}]\right)^{\frac{1}{2p}}h^{\frac{1}{2}}, 1 \leq k \leq N. \tag{3.17}$$

Proof We consider one-step approximations of the SDE (1.1) obtained by the SSBT scheme (3.1)

$$\begin{cases} x^* = x + \theta f(x^*)h, \\ X = x + \dfrac{f(x^*)h + g(x^*)\xi\sqrt{h}}{1 + |f(x^*)|h + |g(x^*)\xi|\sqrt{h}}, \end{cases} \tag{3.18}$$

and by the balanced method presented in [27]

$$\bar{X} = x + \frac{f(x)h + g(x)\xi\sqrt{h}}{1 + |f(x)|h + |g(x)\xi|\sqrt{h}},$$

where components of the d-dimensional vector ξ are Gaussian $\mathcal{N}(0,1)$ i.i.d random variables.

Define

$$X_{t,x}(t+h) = x + \int_t^{t+h} f(X_{t,x}(s))ds + \int_t^{t+h} g(X_{t,x}(s))dW(s)$$

and

$$\rho(t,x) = X_{t,x}(t+h) - X.$$

Then

$$\rho(t,x) = X_{t,x}(t+h) - \bar{X} + \bar{X} - X =: \rho_1(t,x) + \rho_2(t,x). \tag{3.19}$$

The estimations of $\rho_1(t,x)$ are given in [27] as follows.

$$|\mathbb{E}[\rho_1(t,x)]| \leq Kh^{\frac{3}{2}}(1 + |x|^{2\kappa}), \tag{3.20}$$

$$\mathbb{E}[|\rho_1(t,x)|^{2p}] \leq Kh^{2p}(1 + |x|^{4p\kappa}). \tag{3.21}$$

Notice that $\mathbb{E}[\xi] = 0$ and

$$\rho_2(t,x) = \frac{f(x)h + g(x)\xi\sqrt{h}}{1 + |f(x)|h + |g(x)\xi|\sqrt{h}} - \frac{f(x^*)h + g(x^*)\xi\sqrt{h}}{1 + |f(x^*)|h + |g(x^*)\xi|\sqrt{h}},$$

then

$$\begin{aligned}
&|\mathbb{E}[\rho_2(t,x)]| \\
&= \left|\mathbb{E}[\frac{f(x)h}{1 + |f(x)|h + |g(x)\xi|\sqrt{h}} - \frac{f(x^*)h}{1 + |f(x^*)|h + |g(x^*)\xi|\sqrt{h}}]\right| \\
&\leq K\left|\mathbb{E}[h(f(x) - f(x^*)) + h^2(f(x)|f(x^*)| - |f(x)|f(x^*)) + h^{\frac{3}{2}}(f(x)|g(x^*)| - f(x^*)|g(x)|)]\right|.
\end{aligned} \tag{3.22}$$

According to (2.3), (2.5), (2.6), (3.14) and (3.18), it follows that

$$|f(x) - f(x^*)| \leq Kh(1 + |x|^{2\kappa}), \tag{3.23}$$

$$\left|f(x)|f(x^*)| - |f(x)|f(x^*)\right| \leq K(1 + |x|^{2\kappa}), \tag{3.24}$$

$$\left|f(x)|g(x^*)| - f(x^*)|g(x)|\right| \leq K(1 + |x|^{2\kappa}). \tag{3.25}$$

Substituting (3.23)-(3.25) into (3.22), we have

$$|\mathbb{E}[\rho_2(t,x)]| \leq Kh^{\frac{3}{2}}(1 + |x|^{2\kappa}). \tag{3.26}$$

Combining it with (3.20) and (3.19), we obtain

$$|\mathbb{E}[\rho(t,x)]| \leq Kh^{\frac{3}{2}}(1 + |x|^{2\kappa}). \tag{3.27}$$

Similar to the prove of (3.26), we can obtain

$$\mathbb{E}[|\rho_2(t,x)|^{2p}] \leq Kh^{2p}(1 + |x|^{4p\kappa}), \tag{3.28}$$

where we need the inequality $|g(x) - g(x^*)|^2 \leq K(1 + |x|^{2\kappa})$, which can be readily derived from (2.6). Combining (3.28) with (3.21) and (3.19), we have

$$\mathbb{E}[|\rho(t,x)|^{2p}] \leq Kh^{2p}(1 + |x|^{4p\kappa}). \tag{3.29}$$

According to (3.27), (3.29) and the fundamental strong convergence theorem, see Theorem 2.1 in [27], we get that the strong convergence rate of the SSBT scheme (3.1) is $\frac{1}{2}$, that is, the estimation (3.17) holds. □

4. Stability properties

In this section, we discuss the exponential stability of the SSBT scheme (3.1). Some definitions are given firstly.

Definition 4.1 (mean-square contraction). *For a fixed step size h, a numerical method is said to be mean-square contractive if the numerical solutions $X_k,\ k = 0, 1, 2, \ldots,$ satisfy*

$$\mathbb{E}[X_k^T X_k] \leq \mathbb{E}[X_0^T X_0], \quad \forall\ k \geq 0.$$

Definition 4.2 (exponential mean-square stability). *For a fixed step size h, a numerical method is said to be exponentially mean-square stable if the numerical solutions $X_k,\ k = 0, 1, 2, \ldots,$ satisfy*

$$\mathbb{E}[X_k^T X_k] \leq C\mathrm{e}^{-\varsigma t_k}\mathbb{E}[X_0^T X_0], \quad \forall\ k \geq 0,$$

where C and ς are positive constants independent of k.

As a special case of Theorem 5.1 in [3], we give a sufficient condition for exponential stability of the exact solution in mean-square sense.

Lemma 4.1. *If there exist a symmetric, positive-definite matrix $Q_{n\times n}$ and a constant α such that*

$$x^T Q f(x) + \frac{1}{2}\mathrm{trace}\left[g^T(x)Qg(x)\right] \leq \alpha x^T Q x, \tag{4.1}$$

then the solution $X(t)$ of the SDE (1.1) obeys

$$\mathbb{E}[X^T(t)QX(t)] \leq \mathrm{e}^{2\alpha t}\mathbb{E}[X^T(0)QX(0)].$$

Hence, the solution $X(t)$ is exponentially mean-square stable if $\alpha < 0$, and mean-square contractive if $\alpha = 0$.

To consider the exponential mean-square stability of the SSBT scheme (3.1), we need the following estimation.

Lemma 4.2. *Given $l \geq 1,\ h > 0,\ \eta \sim N(0, h)$, it holds that*

$$\mathbb{E}[\frac{1}{1 + l^2 h + l^2|\eta|}] > \frac{1}{(1 + h)l^2\sqrt{\pi}}.$$

Proof

$$
\begin{aligned}
\mathbb{E}[\frac{1}{1+l^2h+l^2|\eta|}] &= \int_{-\infty}^{\infty} \frac{1}{1+l^2h+l^2|x|}\frac{1}{\sqrt{2\pi h}}\mathrm{e}^{-\frac{x^2}{2h}}dx \\
&= \frac{2}{\sqrt{2\pi h}}\int_0^{\infty} \frac{1}{1+l^2h+l^2x}\mathrm{e}^{-\frac{x^2}{2h}}dx \\
&\xlongequal{\text{Let } x=\sqrt{2h}t} \frac{2}{\sqrt{2\pi h}}\int_0^{\infty} \frac{1}{1+l^2h+l^2\sqrt{2h}t}\mathrm{e}^{-t^2}\sqrt{2h}dt \\
&\geq \frac{2}{\sqrt{2\pi h}}\frac{1}{l^2}\int_0^{\infty} \frac{1}{1+h+\sqrt{2h}t}\mathrm{e}^{-t^2}\sqrt{2h}dt \\
&= \frac{1}{1+h}\frac{2}{\sqrt{\pi}}\frac{1}{l^2}\int_0^{\infty} \frac{1}{1+\frac{\sqrt{2h}}{1+h}t}\mathrm{e}^{-t^2}dt \\
&\geq \frac{2}{(1+h)l^2\sqrt{\pi}}\int_0^{\infty} \frac{1}{1+t}\mathrm{e}^{-t^2}dt \\
&= \frac{2}{(1+h)l^2\sqrt{\pi}}\frac{\pi\mathrm{erfi}(1)-\mathrm{Ei}(1)}{2\mathrm{e}} \\
&> \frac{1}{(1+h)l^2\sqrt{\pi}}.
\end{aligned}
$$

In the above inequality we use the fact that

$$
\int_0^{\infty} \frac{1}{1+t}e^{-t^2}dt = \frac{\pi\mathrm{erfi}(1)-\mathrm{Ei}(1)}{2\mathrm{e}} > \frac{1}{2},
$$

where

$$
\mathrm{erfi}(z) = \frac{2}{\sqrt{\pi}}\sum_{n=0}^{\infty}\frac{z^{2n+1}}{n!(2n+1)},\ \mathrm{Ei}(z) = -\lim_{\varepsilon\to 0^+}\left[\int_{-z}^{-\varepsilon}\frac{\mathrm{e}^{-t}}{t}dt + \int_{\varepsilon}^{\infty}\frac{\mathrm{e}^{-t}}{t}dt\right] = \mathrm{P.V.}\int_{-\infty}^{z}\frac{e^t}{t}dt,\ z>0.
$$

□

Theorem 4.3. *Assume the inequality (4.1) holds, then we have the following statements:*

(i) If $\alpha \leq 0$ and $\theta \in [\frac{1}{2}, 1]$, then the SSBT scheme (3.1) is mean-square contractive for all fixed $h > 0$, i.e.,

$$
\mathbb{E}[X_k^T Q X_k] \leq \mathbb{E}[X_0^T Q X_0].
$$

(ii) If $\alpha < 0$ and $\theta \in (\frac{1}{2}, 1]$, then the SSBT scheme (3.1) is exponentially mean-square stable for all fixed $h > 0$.

(iii) If $\alpha < 0$, $\theta \in [0, \frac{1}{2}]$ and $f(x)$ satisfies

$$
f^T(x)Qf(x) \leq Kx^TQx, \tag{4.2}
$$

then there exists a constant $h_0 > 0$ such that the SSBT scheme (3.1) is exponentially mean-square stable for fixed $h \in (0, h_0)$.

Proof Firstly, we prove the statement (i). According to (3.1), the definition of $B(X_k^*, \xi_k)$ and $X_k^T = X_k^{*T} - \theta h f^T(X_k^*)$, we can get

$$
\begin{aligned}
X_{k+1}^T Q X_{k+1} =& X_k^T Q X_k + h^2 f^T(X_k^*)Qf(X_k^*)B^2(X_k^*,\xi_k) + \xi_k^T\sqrt{h}g^T(X_k^*)Qg(X_k^*)\xi_k\sqrt{h}B^2(X_k^*,\xi_k) \\
&+ 2h\big(X_k^{*T}Qf(X_k^*) - \theta h f^T(X_k^*)Qf(X_k^*)\big)B(X_k^*,\xi_k) + 2X_k^TQg(X_k^*)\xi_k\sqrt{h}B(X_k^*,\xi_k) \\
&+ 2hf^T(X_k^*)Qg(X_k^*)\xi_k\sqrt{h}B^2(X_k^*,\xi_k).
\end{aligned}
$$

Then

$$\begin{aligned}\mathbb{E}[X_{k+1}^T QX_{k+1}] \leq & \mathbb{E}[X_k^T QX_k] + (1-2\theta)h^2\mathbb{E}[f^T(X_k^*)Qf(X_k^*)B(X_k^*,\xi_k)] \\ & + 2h\mathbb{E}[X_k^{*T}Qf(X_k^*)B(X_k^*,\xi_k)] + h\mathbb{E}[\text{trace}\left(g^T(X_k^*)Qg(X_k^*)B(X_k^*,\xi_k)\right)],\end{aligned}$$

where we have used $0 < B(X_k^*,\xi_k) < 1$ and

$$\mathbb{E}[\xi_k^T\sqrt{h}g^T(X_k^*)Qg(X_k^*)\xi_k\sqrt{h}B(X_k^*,\xi_k)] = h\mathbb{E}[\text{trace}\left(g^T(X_k^*)Qg(X_k^*)B(X_k^*,\xi_k)\right)].$$

By the condition (4.1) it follows

$$\begin{aligned}\mathbb{E}[X_{k+1}^T QX_{k+1}] \leq & \mathbb{E}[X_k^T QX_k] + (1-2\theta)h^2\mathbb{E}[f^T(X_k^*)Qf(X_k^*)B(X_k^*,\xi_k)] \\ & + 2h\alpha\mathbb{E}[X_k^{*T}QX_k^*B(X_k^*,\xi_k)].\end{aligned} \tag{4.3}$$

If $1-2\theta \leq 0$ and $\alpha \leq 0$, we obtain that

$$\mathbb{E}[X_{k+1}^T QX_{k+1}] \leq \mathbb{E}[X_k^T QX_k].$$

Secondly, we prove the statement (ii). Note that $\alpha < 0$ and $\theta \in (\frac{1}{2}, 1]$. Substituting $hf(X_k^*) = \frac{X_k^* - X_k}{\theta}$ into (4.3) and by Cauchy's inequality, we have

$$\begin{aligned}&\mathbb{E}[X_{k+1}^T QX_{k+1}] \\ &\leq \mathbb{E}[X_k^T QX_k] + \frac{1-2\theta}{\theta^2}\mathbb{E}[\left(X_k^{*T}QX_k^* + X_k^T QX_k - 2X_k^{*T}QX_k\right)B(X_k^*,\xi_k)] \\ &\quad + 2h\alpha\mathbb{E}[X_k^{*T}QX_k^*B(X_k^*,\xi_k)] \\ &= \mathbb{E}[X_k^T QX_k] + \frac{1-2\theta}{\theta^2}\mathbb{E}[X_k^T QX_kB(X_k^*,\xi_k)] + \left(\frac{1-2\theta}{\theta^2} + 2h\alpha\right)\mathbb{E}[X_k^{*T}QX_k^*B(X_k^*,\xi_k)] \\ &\quad + \frac{2\theta-1}{\theta^2}\mathbb{E}[2X_k^{*T}QX_kB(X_k^*,\xi_k)] \\ &\leq \mathbb{E}[X_k^T QX_k] + \frac{1-2\theta}{\theta^2}\mathbb{E}[X_k^T QX_kB(X_k^*,\xi_k)] + \left(\frac{1-2\theta}{\theta^2} + 2h\alpha\right)\mathbb{E}[X_k^{*T}QX_k^*B(X_k^*,\xi_k)] \\ &\quad + \frac{2\theta-1}{\theta^2}\mathbb{E}[\frac{2\theta-1-2\alpha h\theta^2}{2\theta-1}X_k^{*T}QX_k^*B(X_k^*,\xi_k)] + \frac{2\theta-1}{\theta^2}\mathbb{E}[\frac{2\theta-1}{2\theta-1-2\alpha h\theta^2}X_k^T QX_kB(X_k^*,\xi_k)] \\ &= \mathbb{E}[X_k^T QX_k] + \frac{2\alpha h(2\theta-1)}{2\theta-1-2\alpha h\theta^2}\mathbb{E}[X_k^T QX_kB(X_k^*,\xi_k)].\end{aligned} \tag{4.4}$$

Note that $\dfrac{2\theta-1-2\alpha h\theta^2}{2\theta-1} > 0$, which ensures the use of Cauchy's inequality in the above estimation.

Introduce the sets

$$\Omega_l = \{\omega : |f(X_k^*)| \leq l^2 \text{ and } |g(X_k^*)| \leq l^2\} \cap \{\omega : |f(X_k^*)| > (l-1)^2 \text{ or } |g(X_k^*)| > (l-1)^2\},\ l = 1, 2, \ldots$$

Then

$$\begin{aligned}\mathbb{E}[X_k^T QX_kB(X_k^*,\xi_k)] &= \sum_{l=1}^{\infty}\mathbb{E}[\chi_{\Omega_l}X_k^T QX_kB(X_k^*,\xi_k)] \\ &\geq \mathbb{E}[X_k^T QX_k]\sum_{l=1}^{\infty}\mathbb{E}[\frac{1}{1+l^2h+l^2|\xi_k|\sqrt{h}}].\end{aligned}$$

According to Lemma 4.2, we have

$$\mathbb{E}[X_k^T QX_kB(X_k^*,\xi_k)] \geq \mathbb{E}[X_k^T QX_k]\sum_{l=1}^{\infty}\frac{1}{(1+h)l^2\sqrt{\pi}} = \frac{\pi^{\frac{3}{2}}}{6(1+h)}\mathbb{E}[X_k^T QX_k]. \tag{4.5}$$

Substituting (4.5) into (4.4) and noticing that $\alpha < 0$ and $\theta \in (\frac{1}{2}, 1]$, it follows that

$$\begin{aligned}\mathbb{E}[X_{k+1}^T Q X_{k+1}] &\leq \left(1 + \frac{2\alpha h(2\theta - 1)}{2\theta - 1 - 2\alpha h\theta^2} \frac{\pi^{\frac{3}{2}}}{6(1+h)}\right) \mathbb{E}[X_k^T Q X_k] \\ &\leq \exp\left(\frac{2\alpha h(2\theta - 1)}{2\theta - 1 - 2\alpha h\theta^2} \frac{\pi^{\frac{3}{2}}}{6(1+h)}\right) \mathbb{E}[X_k^T Q X_k].\end{aligned}$$

Recursively using the above inequality, we obtain

$$\begin{aligned}\mathbb{E}[X_{k+1}^T Q X_{k+1}] &\leq \exp\left(\frac{\alpha(k+1)h(2\theta - 1)}{2\theta - 1 - 2\alpha h\theta^2} \frac{\pi^{\frac{3}{2}}}{3(1+h)}\right) \mathbb{E}[X_0^T Q X_0] \\ &\leq \mathrm{e}^{\alpha t_{k+1} C(h,\theta)} \mathbb{E}[X_0^T Q X_0].\end{aligned}$$

Therefore the SSBT (3.1) is exponentially mean-square stable $\forall\ h > 0$, and the second statement is proved.

Lastly, we prove the statement (iii). Combining (4.2) and (4.3), it follows

$$\mathbb{E}[X_{k+1}^T Q X_{k+1}] \leq \mathbb{E}[X_k^T Q X_k] + h\left(K(1-2\theta)h + 2\alpha\right) \mathbb{E}[X_k^{*T} Q X_k^* B(X_k^*, \xi_k)]. \tag{4.6}$$

Let

$$h_0 = \begin{cases} +\infty, & \theta = \frac{1}{2}, \\ \frac{-2\alpha}{K(1-2\theta)}, & \theta \in \left[0, \frac{1}{2}\right). \end{cases}$$

By inequalities (4.2), (3.1) and Young's inequality, one has

$$X_k^{*T} Q X_k^* \geq \frac{X_k^T Q X_k}{2(1 + K\theta^2 h^2)}.$$

Therefore, when $h \in (0, h_0)$ the inequality (4.6) yields

$$\mathbb{E}[X_{k+1}^T Q X_{k+1}] \leq \mathbb{E}[X_k^T Q X_k] + h\left(K(1-2\theta)h + 2\alpha\right) \mathbb{E}[\frac{X_k^T Q X_k}{2(1 + K\theta^2 h^2)} B(X_k^*, \xi_k)].$$

Similar to the proof of the second statement, it can be immediately gotten that

$$\begin{aligned}\mathbb{E}[X_{k+1}^T Q X_{k+1}] &\leq \left(1 + \frac{h\left(K(1-2\theta)h + 2\alpha\right)}{2(1 + K\theta^2 h^2)} \frac{\pi^{\frac{3}{2}}}{6(1+h)}\right) \mathbb{E}[X_k^T Q X_k] \\ &\leq \exp\left(\frac{h\left(K(1-2\theta)h + 2\alpha\right)}{2(1 + K\theta^2 h^2)} \frac{\pi^{\frac{3}{2}}}{6(1+h)}\right) \mathbb{E}[X_k^T Q X_k].\end{aligned}$$

Therefore the scheme is exponentially mean-square stable when $h \in (0, h_0)$. □

5. Numerical examples

In this section, we give some numerical examples to show the convergence and stability of the proposed SSBT scheme for solving nonlinear SDEs. The semi-implicit Euler scheme

$$X_{k+1} = X_k + (1-\theta)hf(X_k) + \theta hf(X_{k+1}) + g(X_k)\xi_k\sqrt{h}, \quad \theta \in [0, 1],$$

has been introduced to make comparison. To compute the mean-square error, we run M independent trajectories:

$$\mathrm{Err(h)} := \left(\frac{1}{M}\sum_{i=1}^{M} |X^{Ref}(T, \omega_i) - X_N^{(h)}(\omega_i)|^2\right)^{\frac{1}{2}} \approx \left(\mathbb{E}|X(T) - X_N^{(h)}|^2\right)^{\frac{1}{2}},$$

where $X^{Ref}(T,\omega_i)$ is the reference solution calculated by the SSBT scheme with small step size $h_0 = 2^{-12}$ since it's not easy to obtain the exact solution of the general SDEs.

Example 5.1. *Consider the following nonlinear stochastic differential system*

$$d\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -2x_1 - 3x_1^3 \\ -1.5x_2 - 5x_2^3 \end{bmatrix} dt + \begin{bmatrix} x_1 \sin x_1 & x_1 x_2 \\ x_1 x_2 & 3x_2^2 \end{bmatrix} \begin{bmatrix} dW_1 \\ dW_2 \end{bmatrix}, \tag{5.1}$$

with initial vector
Case I: $X(0) = (x_1(0), x_2(0))^T = (1,1)^T$;
Case II: $X(0) = (x_1(0), x_2(0))^T = (10,10)^T$.

Table 5.1: Comparison of the mean-square errors and convergence rate of the SSBT scheme and the semi-implicit Euler scheme with different values of θ when solving the SDE (5.1). T=1, M=5000 (Example 5.1, **Case I**).

SSBT	$\theta = 0.2$		$\theta = 0.5$		$\theta = 0.7$		$\theta = 1$	
h	Err(h)	rate	Err(h)	rate	Err(h)	rate	Err(h)	rate
2^{-5}	3.80e-02	-	4.41e-02	-	4.99e-02	-	6.02e-02	-
2^{-6}	2.26e-02	0.75	2.46e-02	0.84	2.70e-02	0.89	3.13e-02	0.95
2^{-7}	1.38e-02	0.71	1.41e-02	0.80	1.49e-02	0.85	1.66e-02	0.91
2^{-8}	8.81e-03	0.65	8.64e-03	0.71	8.79e-03	0.76	9.39e-03	0.82
Semi-implicit Euler	$\theta = 0.2$		$\theta = 0.5$		$\theta = 0.7$		$\theta = 1$	
h	Err(h)	rate	Err(h)	rate	Err(h)	rate	Err(h)	rate
2^{-5}	Fail		7.33e-02	-	6.85e-02	-	6.42e-02	-
2^{-6}	Fail		5.33e-02	0.46	5.08e-02	0.43	4.64e-02	0.47
2^{-7}	Fail		3.56e-02	0.58	3.36e-02	0.60	2.91e-02	0.67
2^{-8}	Fail		2.70e-02	0.40	2.55e-02	0.39	2.27e-02	0.36

It is readily verified that the coefficients of the system (5.1) satisfies Assumption 2.1. Recall Theorem 3.3, which states that the convergence rate of the SSBT scheme should be 0.5. In practice, Table 5.1 suggests that when solving the system (5.1) under **Case I**, the convergence rate is greater than 0.5, especially when θ is close to one. In contrast, for $h = 2^{-5}, 2^{-6}, 2^{-7}, 2^{-8}$, by taking the same iteration algorithm for solving the implicit difference scheme as that using for the SSBT scheme, the semi-implicit Euler scheme cannot obtain convergent numerical solution when $\theta = 0.2$, and its accuracy and convergence rates are lower than the SSBT scheme when $\theta = 0.5$, 0.7, and $\theta = 1$, the case of the backward Euler scheme.

According to Lemma 4.1 and direct calculation, we get that the exact solution of system (5.1) is exponential mean-square stable with $\alpha = -1.5$. When $\theta = 0$, the SSBT scheme reduces to the balanced method presented in [27]. Figs. 5.1 and 5.2 show that it is not exponentially stable when taking $h = 1$ for both Cases I and II, while for $\theta = 0.5$, 0.7 and 1, the SSBT scheme with $h = 1$ can preserve the exponential stability of the exact solution of system (5.1) for the two cases. It is also shown by Fig. 5.1 that when taking the initial vector as $X(0) = (1,1)^T$ (Case I), the numerical solution of the semi-implicit Euler scheme with $h = 1$ is exponential mean-square stable in the cases of $\theta = 0.5$, $\theta = 0.7$ and $\theta = 1$. However, the situation is different when taking a larger scale initial vector $X(0) = (10,10)^T$ (Case II). Fig. 5.2 illustrates that the semi-implicit Euler scheme cannot preserve the exponential mean-square stability when $\theta = 0.5$ if taking step size $h = 1$.

It is worth to mention that the system (5.1) does not satisfy the condition (4.2) in Theorem 4.3, while we still observe the exponentially stable numerical solution when $\theta = 0.5$ and $h = 1$. It implies that the inequality (4.2) in the last statement of Theorem 4.3 may not be necessary for $\theta = \frac{1}{2}$.

To further verify the last statement of Theorem 4.3, we consider a toy model in Example 5.2, in which the parameters $\alpha = -8.5$ and $K = 81$ can be calculated directly.

Example 5.2.

$$\begin{cases} dX(t) = -9X(t)dt + X(t)\sin X(t)dW(t), & 0 < t \leq T, \\ X(0) = 1. \end{cases} \tag{5.2}$$

By Theorem 4.3, when $\theta = 0.2$, $h_0 = 0.35$. Fig. 5.3 shows that the numerical solution produced by the SSBT scheme is exponentially mean-square stable when $h = 0.2 < h_0$, while it is not exponentially stable when $h = 0.4 > h_0$. It is well consistent with the theoretical results.

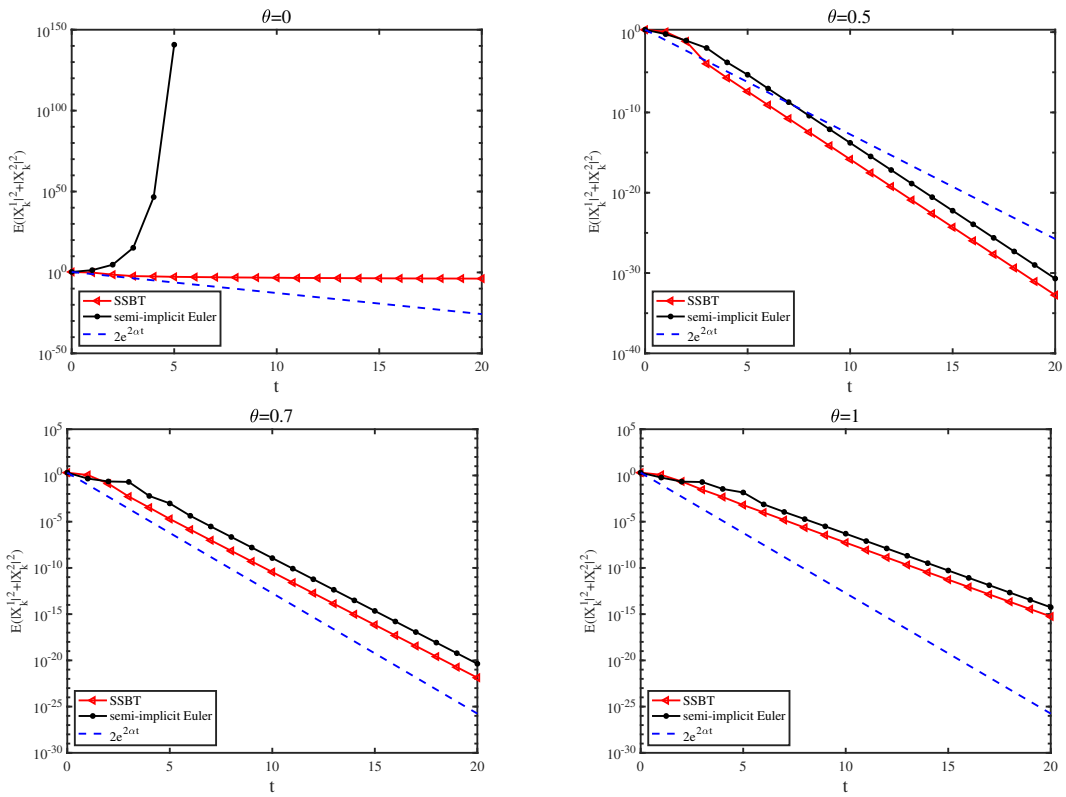

Figure 5.1: Comparison of the $\mathbb{E}(|X_k^1|^2 + |X_k^2|^2)$ of the SSBT scheme and the semi-implicit scheme for solving the SDEs (5.1). $T = 20, h = 1, M = 2000$, the reference line $X_0^T X_0 e^{2\alpha t} = 2e^{-3t}$ (Example 5.1, **Case I**).

6. Conclusions

In this work, we combine the balanced method with the split-step algorithm to construct the SSBT scheme for solving SDE (1.1) under non-globally Lipschitz continuous coefficients. The former method is often adopted to overcome the difficulty caused by nonlinear growth coefficients, and the latter is good at stability. The combination brings us an efficient SSBT scheme (3.1), which is of half-order convergence in mean-square sense. It has been proved in Theorem 4.3 that the SSBT scheme is exponentially mean-square stable when $\theta \in (0.5, 1]$ for every step size h and $\theta \in [0, 0.5]$ for $h < h_0$.

It is numerically shown that compared to the semi-implicit Euler scheme, the SSBT scheme has higher accuracy and convergence rate, as well as better stability behavior for solving (5.1) with both small and large scale initial vectors, see Example 5.1, Case I and II.

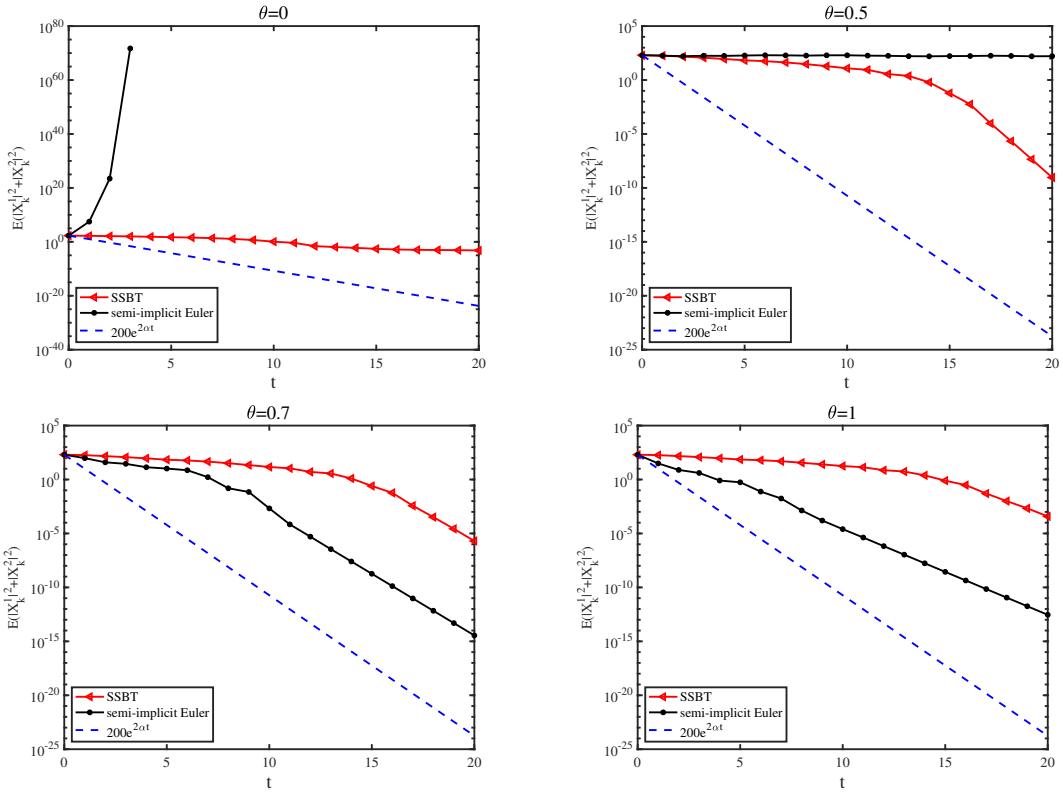

Figure 5.2: Comparison of the $\mathbb{E}(|X_k^1|^2 + |X_k^2|^2)$ of the SSBT scheme and the semi-implicit scheme for solving the SDEs (5.1). $T = 20, h = 1, M = 2000$, the reference line $X_0^T X_0 e^{2\alpha t} = 200e^{-3t}$ (Example 5.1, **Case II**).

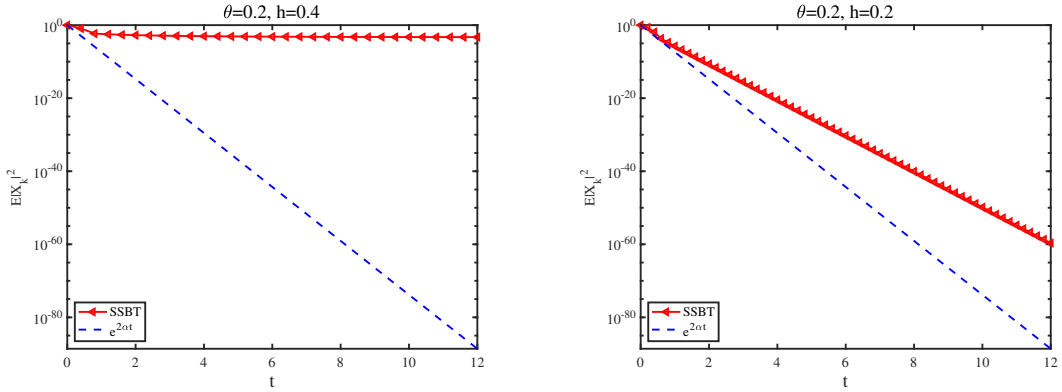

Figure 5.3: Comparison of exponential stability of the SSBT scheme with step size $h > h_0$ and $h < h_0$, $h_0 = 0.35$, $\theta = 0.2$, $T = 12$, $M = 2000$, the reference line is $e^{2\alpha t} = e^{-17t}$ (Example 5.2).

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